Module 2: Process and Process management

A process is a program in execution which then forms the basis of all computation. The process is not as same as program code but a lot more than it. A process is an 'active' entity as opposed to the program which is considered to be a 'passive' entity.

Process Architecture

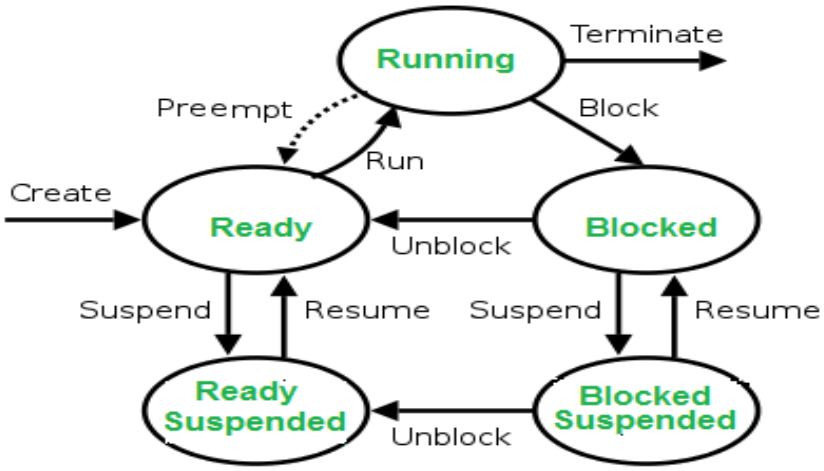
Here, is an Architecture diagram of the Process

- **Stack:** The Stack stores temporary data like function parameters, returns addresses, and local variables.
- **Heap** Allocates memory, which may be processed during its run time.
- **Data:** It contains the variable.
- **Text:** Text Section includes the current activity, which is represented by the value of the Program Counter.

Process States:

A process state is a condition of the process at a specific instant of time. It also defines the current position of the process.

There are mainly seven stages of a process which are:



- **New:** The new process is created when a specific program calls from secondary memory/ hard disk to primary memory/ RAM a
- **Ready:** In a ready state, the process should be loaded into the primary memory, which is ready for execution.
- **Waiting:** The process is waiting for the allocation of CPU time and other resources for execution.
- **Executing:** The process is an execution state.
- **Blocked:** It is a time interval when a process is waiting for an event like I/O operations to complete.
- **Suspended:** Suspended state defines the time when a process is ready for execution but has not been placed in the ready queue by OS.
- **Terminated:** Terminated state specifies the time when a process is terminated

After completing every step, all the resources are used by a process, and memory becomes free.

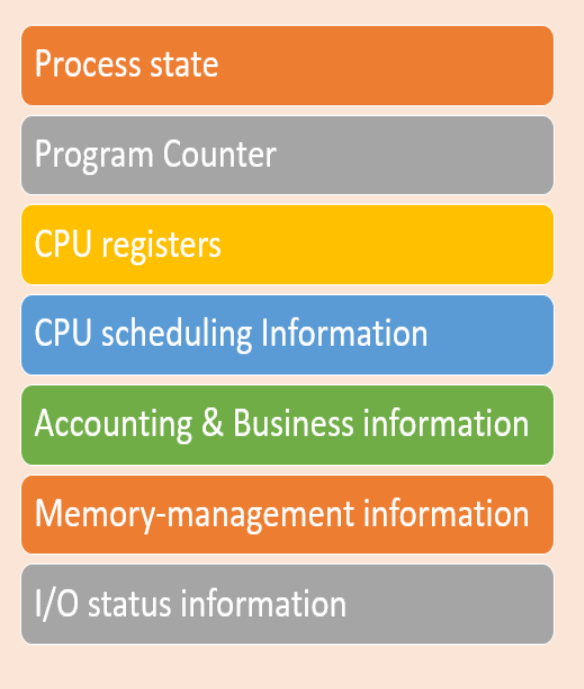
Process Control Block (PCB)

Every process is represented in the operating system by a process control block, which is also called a task control block.

Here, are important components of PCB

- **Process state:** A process can be new, ready, running, waiting, etc.

- **Program counter:** The program counter lets you know the address of the next instruction, which should be executed for that process.
- **CPU registers:** This component includes accumulators, index and general-purpose registers, and information of condition code.
- **CPU scheduling information:** This component includes a process priority, pointers for scheduling queues, and various other scheduling parameters.
- **Accounting and business information:** It includes the amount of CPU and time utilities like real time used, job or process numbers, etc.
- **Memory-management information:** This information includes the value of the base and limit registers, the page, or segment tables. This depends on the memory system, which is used by the operating system.
- **I/O status information:** This block includes a list of open files, the list of I/O devices that are allocated to the process.



What is Process Scheduling?

The act of determining which process is in the **ready** state, and should be moved to the **running** state is known as **Process Scheduling**.

The prime aim of the process scheduling system is to keep the CPU busy all the time and to deliver minimum response time for all programs. For achieving this, the scheduler must apply appropriate rules for swapping processes IN and OUT of CPU.

Scheduling fell into one of the two general categories:

- **Non Pre-emptive Scheduling:** When the currently executing process gives up the CPU voluntarily.
- **Pre-emptive Scheduling:** When the operating system decides to favour another process, pre-empting the currently executing process.

What are Scheduling Queues?

- All processes, upon entering into the system, are stored in the **Job Queue**.
- Processes in the Ready state are placed in the **Ready Queue**.
- Processes waiting for a device to become available are placed in **Device Queues**. There are unique device queues available for each I/O device.

A new process is initially put in the **Ready queue**. It waits in the ready queue until it is selected for execution (or dispatched). Once the process is assigned to the CPU and is executing, one of the following several events can occur:

- The process could issue an I/O request, and then be placed in the **I/O queue**.
- The process could create a new subprocess and wait for its termination.
- The process could be removed forcibly from the CPU, as a result of an interrupt, and be put back in the ready queue.

Types of Schedulers

There are three types of schedulers available:

1. **Long Term Scheduler**
2. **Short Term Scheduler**
3. **Medium Term Scheduler**

Long Term Scheduler

Long term scheduler runs less frequently. Long Term Schedulers decide which program must get into the job queue. From the job queue, the Job Processor, selects processes and loads them into the memory for execution. Primary aim of the Job Scheduler is to maintain a good degree of Multiprogramming. An optimal degree of Multiprogramming means the average rate of process creation is equal to the average departure rate of processes from the execution memory.

Short Term Scheduler

This is also known as CPU Scheduler and runs very frequently. The primary aim of this scheduler is to enhance CPU performance and increase process execution rate.

Medium Term Scheduler

This scheduler removes the processes from memory (and from active contention for the CPU), and thus reduces the degree of multiprogramming. At some later time, the process can be reintroduced into memory and its execution van be continued where it left off. This scheme is called **swapping**. The process is swapped out, and is later swapped in, by the medium term scheduler.

What is Context Switch?

1. Switching the CPU to another process requires **saving** the state of the old process and **loading** the saved state for the new process. This task is known as a **Context Switch**.
2. The **context** of a process is represented in the **Process Control Block(PCB)** of a process; it includes the value of the CPU registers, the process state and memory-management information. When a context switch occurs, the Kernel saves the context of the old process in its PCB and loads the saved context of the new process scheduled to run.
3. Context switch time is **pure overhead**, because the **system does no useful work while switching**. Its speed varies from machine to machine, depending on the memory speed, the number of registers that must be copied, and the existence of special instructions(such as a single instruction to load or store all registers). Typical speeds range from 1 to 1000 microseconds.

4. Context Switching has become such a performance **bottleneck** that programmers are using new structures(threads) to avoid it whenever and wherever possible.

CPU Scheduling: Scheduling Criteria

There are many different criteria to check when considering the **"best"** scheduling algorithm, they are:

CPU Utilization

To make out the best use of the CPU and not to waste any CPU cycle, the CPU would be working most of the time(Ideally 100% of the time). Considering a real system, CPU usage should range from 40% (lightly loaded) to 90% (heavily loaded.)

Throughput

It is the total number of processes completed per unit of time or rather says the total amount of work done in a unit of time. This may range from 10/second to 1/hour depending on the specific processes.

Turnaround Time

It is the amount of time taken to execute a particular process, i.e. The interval from the time of submission of the process to the time of completion of the process(Wall clock time).

Waiting Time

The sum of the periods spent waiting in the ready queue amount of time a process has been waiting in the ready queue to acquire get control on the CPU.

Load Average

It is the average number of processes residing in the ready queue waiting for their turn to get into the CPU.

Response Time

Amount of time it takes from when a request was submitted until the first response is produced. Remember, it is the time till the first response and not the completion of process execution (final response).

In general CPU utilization and Throughput are maximized and other factors are reduced for proper optimization.

Scheduling Algorithms

To decide which process to execute first and which process to execute last to achieve maximum CPU utilization, computer scientists have defined some algorithms, they are:

First Come First Serve Scheduling

In the "First come first serve" scheduling algorithm, as the name suggests,
the process which arrives first, gets executed first, or we can say that the process which requests the CPU first, gets the CPU allocated first.

- First Come First Serve, is just like **FIFO**(First in First out) Queue data structure, where the data element which is added to the queue first, is the one who leaves the queue first.
- This is used in Batch Systems.
- It's **easy to understand and implement** programmatically, using a Queue data structure, where a new process enters through the **tail** of the queue, and the scheduler selects process from the **head** of the queue.
- A perfect real life example of FCFS scheduling is **buying tickets at ticket counter**.

Calculating Average Waiting Time

For every scheduling algorithm, **Average waiting time** is a crucial parameter to judge it's performance.

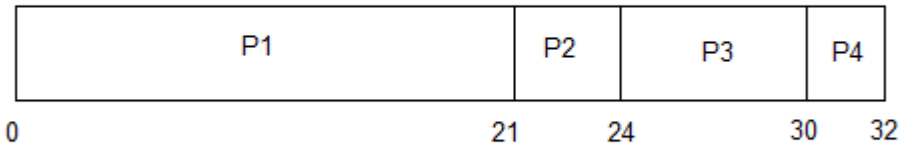
AWT or Average waiting time is the average of the waiting times of the processes in the queue, waiting for the scheduler to pick them for execution.

Lower the Average Waiting Time, better the scheduling algorithm.

Consider the processes P1, P2, P3, P4 given in the below table, arrives for execution in the same order, with **Arrival Time** 0, and given **Burst Time**, let's find the average waiting time using the FCFS scheduling algorithm.

PROCESS	BURST TIME
P1	21
P2	3
P3	6
P4	2

The average waiting time will be = (0 + 21 + 24 + 30)/4 = 18.75 ms



This is the GANTT chart for the above processes

The average waiting time will be 18.75 ms

For the above given proccesses, first **P1** will be provided with the CPU resources,

- Hence, waiting time for **P1** will be 0
- **P1** requires 21 ms for completion, hence waiting time for **P2** will be 21 ms
- Similarly, waiting time for process **P3** will be execution time of **P1** + execution time for **P2**, which will be (21 + 3) ms = 24 ms.
- For process **P4** it will be the sum of execution times of **P1**, **P2** and **P3**.

The **GANTT chart** above perfectly represents the waiting time for each process

Problems with FCFS Scheduling

Below we have a few shortcomings or problems with the FCFS scheduling algorithm:

1. It is **Non Pre-emptive** algorithm, which means the **process priority** doesn't matter.

   If a process with very least priority is being executed, more like **daily routine backup** process, which takes more time, and all of a sudden some other high priority process arrives, like **interrupt to avoid system crash**, the high priority process will have to wait, and hence in this case, the system will crash, just because of improper process scheduling.

2. Not optimal Average Waiting Time.
3. Resources utilization in parallel is not possible, which leads to **Convoy Effect**, and hence poor resource(CPU, I/O etc) utilization

Shortest Job First(SJF) Scheduling

Shortest Job First scheduling works on the process with the shortest **burst time** or **duration** first.

- This is the best approach to minimize waiting time.
- This is used in Batch Systems.
- It is of two types:
    1. Non Pre-emptive
    2. Pre-emptive
- To successfully implement it, the burst time/duration time of the processes should be known to the processor in advance, which is practically not feasible all the time.
- This scheduling algorithm is optimal if all the jobs/processes are available at the same time. (either Arrival time is `0` for all, or Arrival time is same for all)

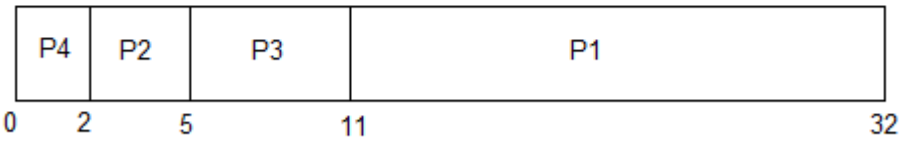
Non Pre-emptive Shortest Job First

Consider the below processes available in the ready queue for execution, with **arrival time** as `0` for all and given **burst times**.

PROCESS	BURST TIME
P1	21
P2	3
P3	6
P4	2

In Shortest Job First Scheduling, the shortest Process is executed first.

Hence the GANTT chart will be following :



Now, the average waiting time will be = (0 + 2 + 5 + 11)/4 = 4.5 ms

As you can see in the **GANTT chart** above, the process **P4** will be picked up first as it has the shortest burst time, then **P2**, followed by **P3** and at last **P1**.

We scheduled the same set of processes using the First come first serve algorithm in the previous tutorial, and got average waiting time to be 18.75 ms, whereas with SJF, the average waiting time comes out 4.5 ms.

Problem with Non Pre-emptive SJF

If the **arrival time** for processes are different, which means all the processes are not available in the ready queue at time 0, and some jobs arrive after some time, in such situation, sometimes process with short burst time have to wait for the current process's execution to finish, because in Non Pre-emptive SJF, on arrival of a process with short duration, the existing job/process's execution is not halted/stopped to execute the short job first.

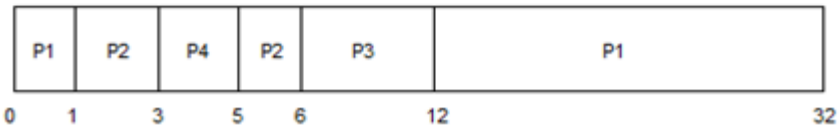
This leads to the problem of **Starvation**, where a shorter process has to wait for a long time until the current longer process gets executed. This happens if shorter jobs keep coming, but this can be solved using the concept of **aging**.

Pre-emptive Shortest Job First

In Preemptive Shortest Job First Scheduling, jobs are put into ready queue as they arrive, but as a process with **short burst time** arrives, the existing process is preempted or removed from execution, and the shorter job is executed first.

PROCESS	BURST TIME	ARRIVAL TIME
P1	21	0
P2	3	1
P3	6	2
P4	2	3

The GANTT chart for Preemptive Shortest Job First Scheduling will be,



The average waiting time will be,((5-3)+(6-2)+(12-1))/4=8.75

The average waiting time for preemptive shortest job first scheduling is less than both,non preemptive SJF scheduling and FCFS scheduling

As you can see in the **GANTT chart** above, as **P1** arrives first, hence it's execution starts immediately, but just after 1 ms, process **P2** arrives with a **burst time** of 3 ms which is less than the burst time of **P1**, hence the process **P1**(1 ms done, 20 ms left) is preemptied and process **P2** is executed.

As **P2** is getting executed, after 1 ms, **P3** arrives, but it has a burst time greater than that of **P2**, hence execution of **P2** continues. But after another millisecond, **P4** arrives with a burst time of 2 ms, as a result **P2**(2 ms done, 1 ms left) is preemptied and **P4** is executed.

After the completion of **P4**, process **P2** is picked up and finishes, then **P2** will get executed and at last **P1**.

The Pre-emptive SJF is also known as **Shortest Remaining Time First**, because at any given point of time, the job with the shortest remaining time is executed first.

Priority CPU Scheduling

In the Shortest Job First scheduling algorithm, the priority of a process is generally the inverse of the CPU burst time, i.e. the larger the burst time the lower is the priority of that process.

In case of priority scheduling the priority is not always set as the inverse of the CPU burst time, rather it can be internally or externally set, but yes the scheduling is done on the basis of priority of the process where the process which is most urgent is processed first, followed by the ones with lesser priority in order.

Processes with same priority are executed in FCFS manner.

The priority of process, when internally defined, can be decided based on **memory requirements**, **time limits** ,**number of open files**, **ratio of I/O burst to CPU burst** etc.

Whereas, external priorities are set based on criteria outside the operating system, like the importance of the process, funds paid for the computer resource use, makrte factor etc.

Types of Priority Scheduling Algorithm

Priority scheduling can be of two types:

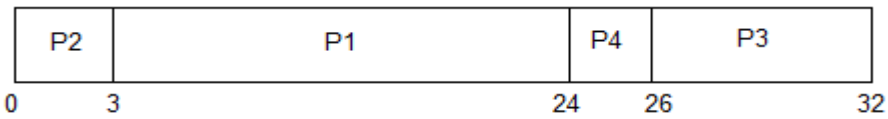
1. **Preemptive Priority Scheduling**: If the new process arrived at the ready queue has a higher priority than the currently running process, the CPU is preempted, which means the processing of the current process is stoped and the incoming new process with higher priority gets the CPU for its execution.
2. **Non-Preemptive Priority Scheduling**: In case of non-preemptive priority scheduling algorithm if a new process arrives with a higher priority than the current running process, the incoming process is put at the head of the ready queue, which means after the execution of the current process it will be processed.

Example of Priority Scheduling Algorithm

Consider the below table fo processes with their respective CPU burst times and the priorities.

PROCESS	BURST TIME	PRIORITY
P1	21	2
P2	3	1
P3	6	4
P4	2	3

The GANTT chart for following processes based on Priority scheduling will be,



The average waiting time will be, (0 + 3 + 24 + 26)/4 = 13.25 ms

As you can see in the GANTT chart that the processes are given CPU time just on the basis of the priorities.

Problem with Priority Scheduling Algorithm

In priority scheduling algorithm, the chances of **indefinite blocking** or **starvation**.

A process is considered blocked when it is ready to run but has to wait for the CPU as some other process is running currently.

But in case of priority scheduling if new higher priority processes keeps coming in the ready queue then the processes waiting in the ready queue with lower priority may have to wait for long durations before getting the CPU for execution.

Using Aging Technique with Priority Scheduling

To prevent starvation of any process, we can use the concept of **aging** where we keep on increasing the priority of low-priority process based on the its waiting time.

For example, if we decide the aging factor to be **0.5** for each day of waiting, then if a process with priority **20**(which is comparitively low priority) comes in the ready queue. After one day of waiting, its priority is increased to **19.5** and so on.

Doing so, we can ensure that no process will have to wait for indefinite time for getting CPU time for processing.

Round Robin Scheduling

Round Robin(RR) scheduling algorithm is mainly designed for time-sharing systems. This algorithm is similar to FCFS scheduling, but in Round Robin(RR) scheduling, preemption is added which enables the system to switch between processes.

- A fixed time is allotted to each process, called a **quantum**, for execution.
- Once a process is executed for the given time period that process is preempted and another process executes for the given time period.
- Context switching is used to save states of preempted processes.
- This algorithm is simple and easy to implement and the most important is thing is this algorithm is starvation-free as all processes get a fair share of CPU.
- It is important to note here that the length of time quantum is generally from 10 to 100 milliseconds in length.

Some important characteristics of the Round Robin(RR) Algorithm are as follows:

1. Round Robin Scheduling algorithm resides under the category of Preemptive Algorithms.
2. This algorithm is one of the oldest, easiest, and fairest algorithm.
3. This Algorithm is a real-time algorithm because it responds to the event within a specific time limit.
4. In this algorithm, the time slice should be the minimum that is assigned to a specific task that needs to be processed. Though it may vary for different operating systems.
5. This is a hybrid model and is clock-driven in nature.

6. This is a widely used scheduling method in the traditional operating system.

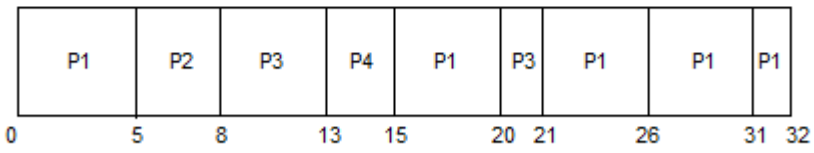
Important terms

1. **Completion Time**
   It is the time at which any process completes its execution.
2. **Turn Around Time**
   This mainly indicates the time Difference between completion time and arrival time. The Formula to calculate the same is: **Turn Around Time = Completion Time – Arrival Time**
3. **Waiting Time(W.T):**
   It Indicates the time Difference between turn around time and burst time.
   And is calculated as **Waiting Time = Turn Around Time – Burst Time**

Let us now cover an example for the same:

PROCESS	BURST TIME
P1	21
P2	3
P3	6
P4	2

The GANTT chart for round robin scheduling will be,



The average waiting time will be, 11 ms.

In the above diagram, arrival time is not mentioned so it is taken as 0 for all processes.

Note: If arrival time is not given for any problem statement then it is taken as 0 for all processes; if it is given then the problem can be solved accordingly.

Explanation

The value of time quantum in the above example is 5.Let us now calculate the Turn around time and waiting time for the above example :

Processes	**Burst Time**	**Turn Around Time** **Turn Around Time = Completion Time – Arrival Time**	**Waiting Time** **Waiting Time = Turn Around Time – Burst Time**
P1	21	32-0=32	32-21=11
P2	3	8-0=8	8-3=5
P3	6	21-0=21	21-6=15
P4	2	15-0=15	15-2=13

Average waiting time is calculated by adding the waiting time of all processes and then dividing them by no.of processes.

average waiting time = waiting time of all processes/ no.of processes

average waiting time=11+5+15+13/4 = 44/4= **11ms**

What are Threads?

Thread is an execution unit that consists of its own program counter, a stack, and a set of registers where the program counter mainly keeps track of which instruction to execute next, a set of registers mainly hold its current working variables, and a stack mainly contains the history of execution

Threads are also known as Lightweight processes. Threads are a popular way to improve the performance of an application through parallelism. Threads are mainly

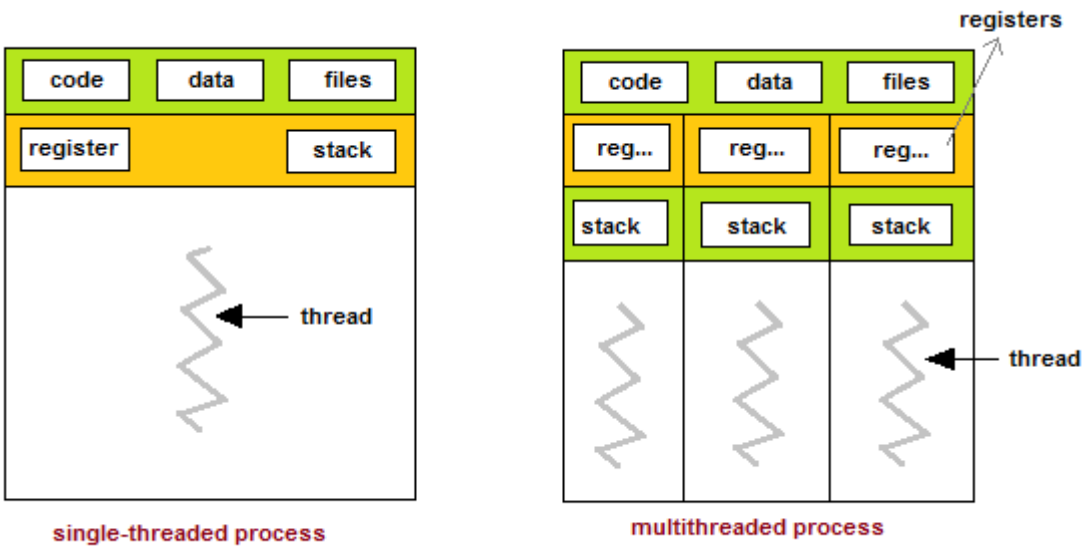
used to represent a software approach in order to improve the performance of an operating system just by reducing the overhead thread that is mainly equivalent to a classical process.

The CPU switches rapidly back and forth among the threads giving the illusion that the threads are running in parallel.

As each thread has its own independent resource for process execution; thus Multiple processes can be executed parallelly by increasing the number of threads.

It is important to note here that each thread belongs to exactly one process and outside a process no threads exist. Each thread basically represents the flow of control separately. In the implementation of network servers and web servers threads have been successfully used. Threads provide a suitable foundation for the parallel execution of applications on shared-memory multiprocessors.

The given below figure shows the working of a single-threaded and a multithreaded process:



Before moving on further let us first understand the difference between a process and a thread.

Process	Thread

A Process simply means any program in execution.	Thread simply means a segment of a process.
The process consumes more resources	Thread consumes fewer resources.
The process requires more time for creation.	Thread requires comparatively less time for creation than process.
The process is a heavyweight process	Thread is known as a lightweight process
The process takes more time to terminate	The thread takes less time to terminate.
Processes have independent data and code segments	A thread mainly shares the data segment, code segment, files, etc. with its peer threads.
The process takes more time for context switching.	The thread takes less time for context switching.
Communication between processes needs more time as compared to thread.	Communication between threads needs less time as compared to processes.
For some reason, if a process gets blocked then the remaining processes can continue their execution	In case if a user-level thread gets blocked, all of its peer threads also get blocked.

Advantages of Thread

Some advantages of thread are given below:

1. Responsiveness
2. Resource sharing, hence allowing better utilization of resources.
3. Economy. Creating and managing threads becomes easier.
4. Scalability. One thread runs on one CPU. In Multithreaded processes, threads can be distributed over a series of processors to scale.
5. Context Switching is smooth. Context switching refers to the procedure followed by the CPU to change from one task to another.
6. Enhanced Throughput of the system. Let us take an example for this: suppose a process is divided into multiple threads, and the function of each thread is considered as one job, then the number of jobs completed per unit of time increases which then leads to an increase in the throughput of the system.

Types of Thread

There are two types of threads:

1. User Threads
2. Kernel Threads

User threads are above the kernel and without kernel support. These are the threads that application programmers use in their programs.

Kernel threads are supported within the kernel of the OS itself. All modern OSs support kernel-level threads, allowing the kernel to perform multiple simultaneous tasks and/or to service multiple kernel system calls simultaneously.

Let us now understand the basic difference between User level Threads and Kernel level threads:

User Level threads	**Kernel Level Threads**
These threads are implemented by users.	These threads are implemented by Operating

	systems
These threads are not recognized by operating systems,	These threads are recognized by operating systems,
In User Level threads, the Context switch requires no hardware support.	In Kernel Level threads, hardware support is needed.
These threads are mainly designed as dependent threads.	These threads are mainly designed as independent threads.
In User Level threads, if one user-level thread performs a blocking operation then the entire process will be blocked.	On the other hand, if one kernel thread performs a blocking operation then another thread can continue the execution.
Example of User Level threads: Java thread, POSIX threads.	Example of Kernel level threads: Window Solaris.
Implementation of User Level thread is done by a thread library and is easy.	While the Implementation of the kernel-level thread is done by the operating system and is complex.

Multithreading Models

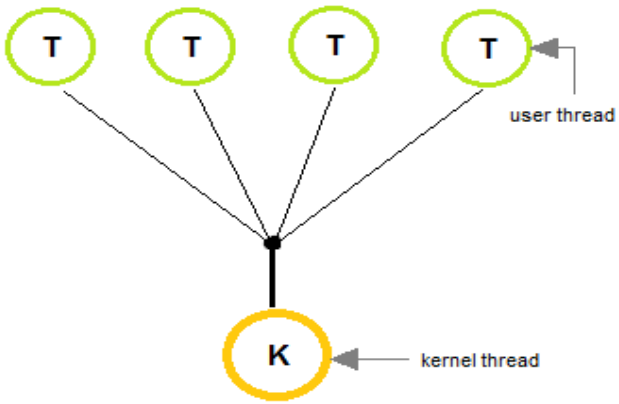
The user threads must be mapped to kernel threads, by one of the following strategies:

- Many to One Model
- One to One Model

- Many to Many Model

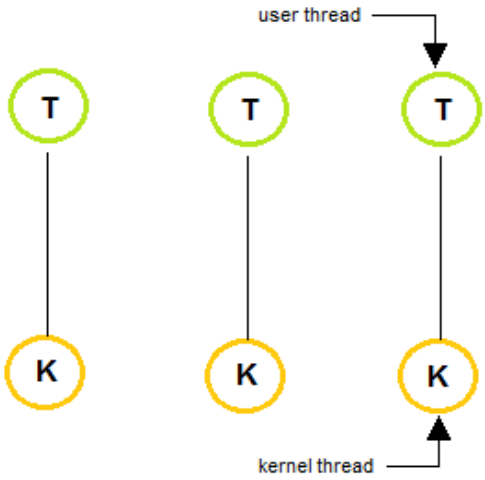
Many to One Model

- In the **many to one** model, many user-level threads are all mapped onto a single kernel thread.
- Thread management is handled by the thread library in user space, which is efficient in nature.
- In this case, if user-level thread libraries are implemented in the operating system in some way that the system does not support them, then the Kernel threads use this many-to-one relationship model.



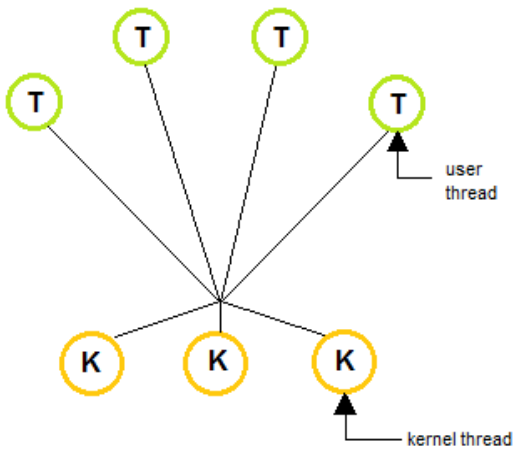
One to One Model

- The **one to one** model creates a separate kernel thread to handle each and every user thread.
- Most implementations of this model place a limit on how many threads can be created.
- Linux and Windows from 95 to XP implement the one-to-one model for threads.
- This model provides more concurrency than that of many to one Model.



Many to Many Model

- The **many to many** model multiplexes any number of user threads onto an equal or smaller number of kernel threads, combining the best features of the one-to-one and many-to-one models.
- Users can create any number of threads.
- Blocking the kernel system calls does not block the entire process.
- Processes can be split across multiple processors.



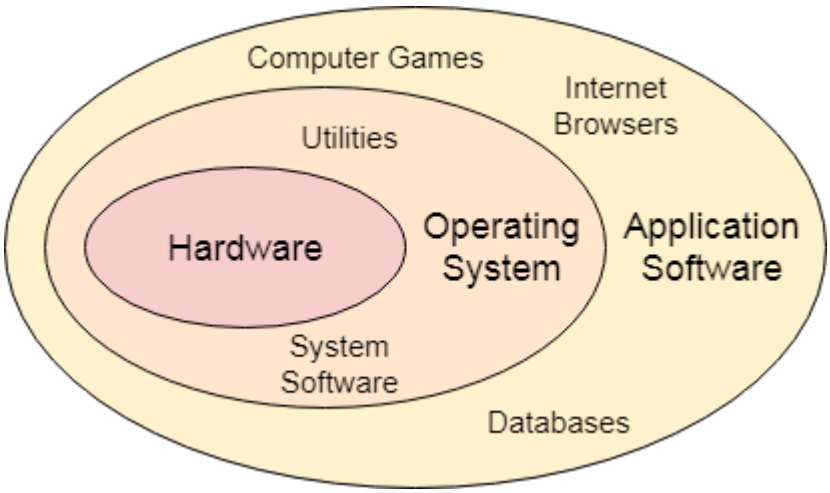
Module 1: Operating system Overview

- 1.1 Introduction, Objectives, Functions and Evolution of Operating System
- 1.2 Operating system structures: Layered, Monolithic and Microkernel
- 1.3 Linux Kernel, Shell and System Calls

What is an operating System?

In the Computer System (comprises of Hardware and software), Hardware can only understand machine code (in the form of 0 and 1) which doesn't make any sense to a naive user.

We need a system which can act as an intermediary and manage all the processes and resources present in the system.



An **Operating System** can be defined as an **interface between user and hardware**. It is responsible for the execution of all the processes, Resource Allocation, CPU management, File Management and many other tasks.

The purpose of an operating system is to provide an environment in which a user can execute programs in convenient and efficient manner.

The operating system is a system program that serves as an interface between the computing system and the end-user. Operating systems create an environment where the user can run any programs or communicate with software or applications in a comfortable and well-organized way.

Furthermore, an operating is a software program that manages and controls the execution of application programs, software resources and computer hardware. It also helps manage the software/hardware resource, such as file management, memory management, input/ output and many peripheral devices like a disk drive, printers, etc.

- ***Explain different objectives of operating System?***

There are three objectives of operating system:

A) **Convenience**: An OS makes a computer more convenient to use.
B) **Efficiency**: An OS allows the computer system resources to be used in an efficient manner.
C) **Ability to evolve**: An OS should be constructed in such a way as to permit the effective development, testing and introduction of new system functions at the same time without interfering with service.

- ***Explain different Functions of operating System?***

The most important functions of operating system are:

A) Process Management: The operating system is responsible for the following activities in connection with the process management

- Scheduling processes and threads on the CPU
- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanism for process synchronization
- Providing mechanisms for process communication

B) Memory Management: The operating system is responsible for the following activities in connection with the memory management

- Keeping track of which parts of memory is currently being used and by whom
- Deciding which processes and data to move into and out of memory
- Allocating and de allocating memory space as needed.

C) File Management: The operating system is responsible for the following activities in connection with the file management

- Creating and deleting files
- Creating and deleting directories to organize files

- Supporting primitives for manipulating files and directories
- Mapping files onto secondary storage
- Backing up the files on non-volatile storage media

D) Device Management: The operating system is responsible for the following activities in connection with the device management

- Keeps tracks of all devices connected to system.
- designates a program responsible for every device known as the Input/Output controller
- Decides which process gets access to a certain device and for how long.
- Allocates devices in an effective and efficient way.
- De allocates devices when they are no longer required.

E) Security: The operating system uses password protection to protect user data and similar other techniques. it also prevents unauthorized access to programs and user data.

F) Booting of computer: The operating system boots the computer.

G) Control over system performance –

Monitors overall system health to help improve performance.

- Records the response time between service requests and system response to have a complete view of the system health.

- *Explain different types of operating System?*

A) Simple Batch Systems

- In this type of system, there is **no direct interaction between user and the computer**.
- The user has to submit a job (written on cards or tape) to a computer operator.
- Then computer operator places a batch of several jobs on an input device.
- Jobs are batched together by type of languages and requirement.
- Then a special program, the monitor, manages the execution of each program in the batch.
- The monitor is always in the main memory and available for execution.

B) Multiprogramming Operating System

- Multiprogramming is an extension to the batch processing where the CPU is kept always busy.
- Each process needs two types of system time: CPU time and IO time.
- In multiprogramming environment, for the time a process does its I/O, The CPU can start the execution of other processes. Therefore, multiprogramming improves the efficiency of the system.

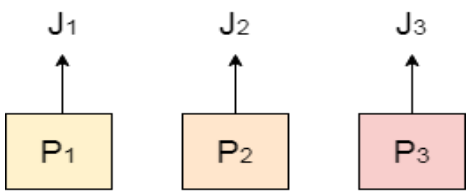
- In Non-multiprogrammed system, there are moments when CPU sits idle and does not do any work.
- In Multiprogramming system, CPU will never be idle and keeps on processing.

Time Sharing Systems are very similar to Multiprogramming batch systems. In fact time sharing systems are an extension of multiprogramming systems.

In Time sharing systems the prime focus is on **minimizing the response time**, while in multiprogramming the prime focus is to maximize the CPU usage.

C) Multiprocessing Operating System

- In Multiprocessing, Parallel computing is achieved. There are more than one processors present in the system which can execute more than one process at the same time. This will increase the throughput of the system.



Multi Processing

D) Real Time Operating System

- In Real Time systems, each job carries a certain deadline within which the Job is supposed to be completed, otherwise the huge loss will be there or even if the result is produced then it will be completely useless.
- The Application of a Real Time system exists in the case of military applications, if you want to drop a missile then the missile is supposed to be dropped with certain precision.

E) Distributed Operating System

- The motivation behind developing distributed operating systems is the availability of powerful and inexpensive microprocessors and advances in communication technology.
- These advancements in technology have made it possible to design and develop distributed systems comprising of many computers that are inter connected by communication networks. The main benefit of distributed systems is its low price/performance ratio.

F) Clustered Systems

- Like parallel systems, clustered systems gather together multiple CPUs to accomplish computational work.
- Clustered systems differ from parallel systems, however, in that they are composed of two or more individual systems coupled together.
- The definition of the term clustered is **not concrete;** the general accepted definition is that clustered computers share storage and are closely linked via LAN networking.
- Clustering is usually performed to provide **high availability**.

G) Handheld Systems

- Handheld systems include **Personal Digital Assistants(PDAs)**, such as Cellular Telephones with connectivity to a network such as the Internet. They are usually of limited size due to which most handheld devices have a small amount of memory, include slow processors, and feature small display screens.

- ***Explain services provided by operating System?***

An Operating System supplies different kinds of services to both the users and to the programs as well. It also provides application programs (that run within an Operating system) an environment to execute it freely.

Here is a list of common services offered by an almost all operating systems:

A) **User Interface**Usually Operating system comes in three forms or types. Depending on the interface their types have been further subdivided. These are:

- Command line interface
- Batch based interface
- Graphical User Interface

The command line interface (CLI) usually deals with using text commands and a technique for entering those commands. The batch interface (BI): commands and directives are used to manage those commands that are entered into files and those files get executed. Another type is the graphical user interface (GUI): which is a window system with a pointing device (like mouse or trackball) to point to the I/O, choose from menus driven interface and to make choices viewing from a number of lists and a keyboard to entry the texts.

B) **Program Execution**

The operating system must have the capability to load a program into memory and execute that program. Furthermore, the program must be able to end its execution, either normally or abnormally / forcefully.

C) **File system manipulation**

Programs need has to be read and then write them as files and directories. File handling portion of operating system also allows users to create and delete files by specific name along with extension, search for a given file and / or list file information. Some programs comprise of permissions management for allowing or denying access to files or directories based on file ownership.

D) **Input / Output Operations**

A program which is currently executing may require I/O, which may involve file or other I/O device. For efficiency and protection, users cannot directly govern the I/O devices. So, the OS provide a means to do I/O Input / Output operation which means read or write operation with any file.

E) Communication

Process needs to swap over information with other process. Processes executing on same computer system or on different computer systems can communicate using operating system support. Communication between two processes can be done using shared memory or via message passing.

F) Resource Allocation

When multiple jobs running concurrently, resources must need to be allocated to each of them. Resources can be CPU cycles, main memory storage, file storage and I/O devices. CPU scheduling routines are used here to establish how best the CPU can be used.

G) Error Detection

Errors may occur within CPU, memory hardware, I/O devices and in the user program. For each type of error, the OS takes adequate action for ensuring correct and consistent computing.

H) Accounting

This service of the operating system keeps track of which users are using how much and what kinds of computer resources have been used for accounting or simply to accumulate usage statistics.

I) Security and protection

Protection includes in ensuring all access to system resources in a controlled manner. For making a system secure, the user needs to authenticate him or her to the system before using (usually via login ID and password).

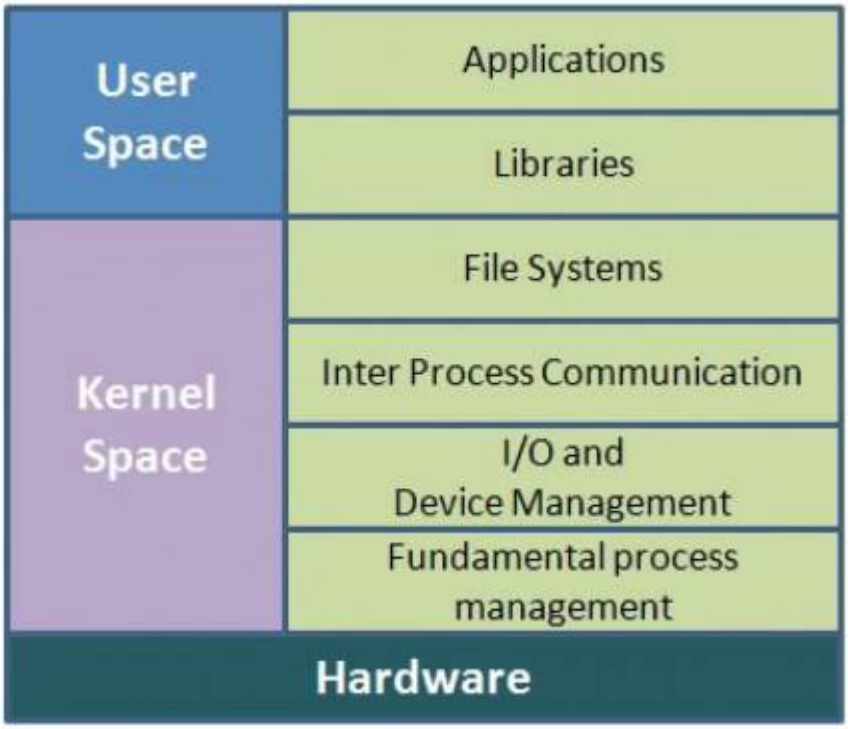
Explain the operating System Structure

The operating system structure are categorized as-

A)Monolithic Approach

The monolithic operating system is also known as the monolithic kernel. This is an old type of operating system. They were used to perform small tasks like batch processing, time sharing tasks in banks. Monolithic kernel acts as a virtual machine which controls all hardware parts. It is different than microkernel which has limited tasks.
A **microkernel** is divided into two parts i.e. kernel space and user space. Both these parts communicate with each other through IPC (Inter-process communication). Microkernel advantage is that if one server fails then other server takes control of it.
Operating systems which use monolithic architecture were first time used in the 1970's.

The monolithic operating system is a very basic operating system in which file management, memory management, device management, and process management is directly controlled within the kernel. All these components like file management, memory management etc. are located within the kernel.



Advantages of Monolithic Kernel –

- One of the major advantage of having monolithic kernel is that it provides CPU scheduling, memory management, file management and other operating system functions through system calls.
- The other one is that it is a single large process running entirely in a single address space.
- It is a single static binary file. Example of some Monolithic Kernel based OSs are: Unix, Linux, Open VMS, XTS-400, z/TPF.

Disadvantages of Monolithic Kernel –

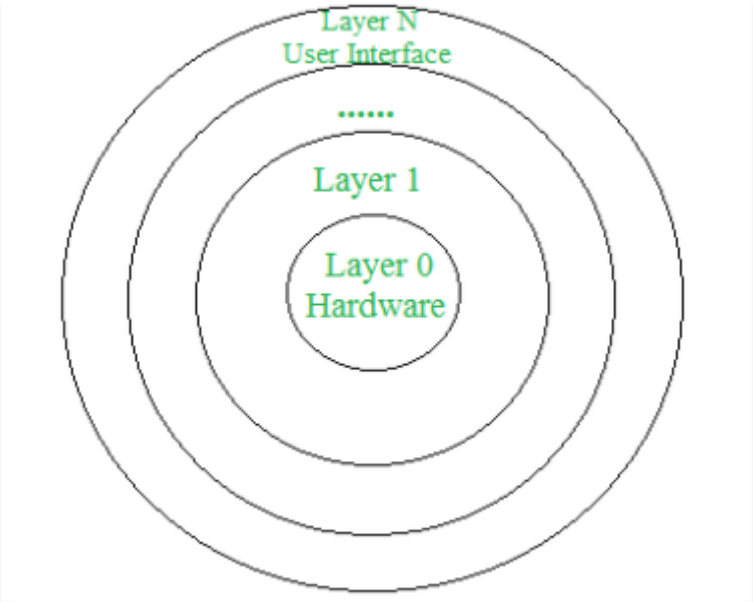
- One of the major disadvantage of monolithic kernel is that, if anyone service fails it leads to entire system failure.
- If user has to add any new service. User needs to modify entire operating system.

B) Layered Approach

An OS can be broken into pieces and retain much more control on system. In this structure the OS is broken into number of layers (levels). The bottom layer (layer 0) is the hardware and the topmost layer (layer N) is the user interface. These layers are so designed that each layer uses the functions of the lower level layers only. This simplifies

the debugging process as if lower level layers are debugged and an error occurs during debugging then the error must be on that layer only as the lower level layers have already been debugged.

The main disadvantage of this structure is that at each layer, the data needs to be modified and passed on which adds overhead to the system. Moreover careful planning of the layers is necessary as a layer can use only lower level layers. UNIX is an example of this structure.



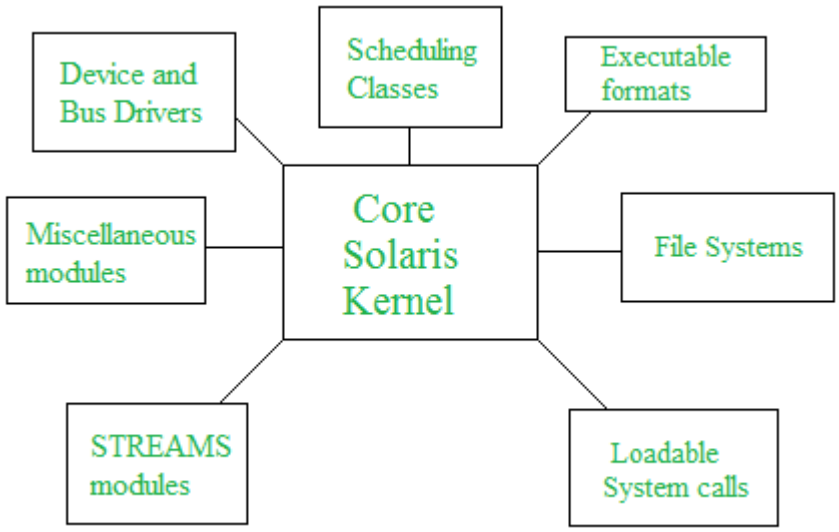
C) Microkernel approach

This structure designs the operating system by removing all non-essential components from the kernel and implementing them as system and user programs. This result in a smaller kernel called the micro-kernel.
Advantages of this structure are that all new services need to be added to user space and does not require the kernel to be modified. Thus it is more secure and reliable as if a service fails then rest of the operating system remains untouched. Mac OS is an example of this type of OS.

Modular structure or approach:
It is considered as the best approach for an OS. It involves designing of a modular kernel. The kernel has only set of core components and other services are added as dynamically loadable modules to the kernel either during run time or boot time. It resembles layered structure due to the fact that each kernel has defined and protected

interfaces but it is more flexible than the layered structure as a module can call any other module.

For example Solaris OS is organized as shown in the figure.



Explain the differences between monolithic and microkernel System Structure

Basis of Comparison	**Monolithic Kernel**	**MicroKernel**
Execution Style	All processes are executed under the kernel space in privileged mode.	Only the most important processes take place in the Kernel space. All other processes are executed in the user space.
Size	Kernel size is bigger when compared to Microkernel.	Kernel size is smaller with respect to the monolithic kernel.
Speed	It provides faster execution of processes.	Process execution is slower.

Stability	A single process crash will cause the entire system to crash.	A single process crash will have no impact on other processes.
Inter-Process Communication	Use signals and sockets to achieve interprocess communication.	Use messaging queues to achieve inter process communication.
Extensibility	Difficult to extend.	Easily extensible.
Maintainability	Maintenance is more time and resource consuming.	Easily maintainable
Debug	Harder to debug	Easier to debug
Security	Less Secure.	More Secure
Example	Linux	Mac OS

System Calls

To understand system calls, first one needs to understand the difference between **kernel mode** and **user mode** of a CPU. Every modern operating system supports these two modes.

Modes supported by the operating system

Kernel Mode

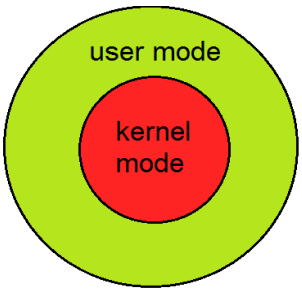
- When CPU is in **kernel mode**, the code being executed can access any memory address and any hardware resource.
- Hence kernel mode is a very privileged and powerful mode.

- If a program crashes in kernel mode, the entire system will be halted.

User Mode

- When CPU is in **user mode**, the programs don't have direct access to memory and hardware resources.
- In user mode, if any program crashes, only that particular program is halted.
- That means the system will be in a safe state even if a program in user mode crashes.
- Hence, most programs in an OS run in user mode.

When a program in user mode requires access to RAM or a hardware resource, it must ask the kernel to provide access to that resource. This is done via something called a **system call**.



When a program makes a system call, the mode is switched from user mode to kernel mode. This is called a **context switch**.

Then the kernel provides the resource which the program requested. After that, another context switch happens which results in change of mode from kernel mode back to user mode.

A **system call** is a mechanism that provides the interface between a process and the operating system. It is a programmatic method in which a computer program requests a service from the kernel of the OS.

System call offers the services of the operating system to the user programs via API (Application Programming Interface). System calls are the only entry points for the kernel system.

Why do you need System Calls in OS?

Following are situations which need system calls in OS:

- Reading and writing from files demand system calls.
- If a file system wants to create or delete files, system calls are required.
- System calls are used for the creation and management of new processes.
- Network connections need system calls for sending and receiving packets.
- Access to hardware devices like scanner, printer, need a system call.

Types of System calls

Here are the five types of system calls used in OS:

- Process Control
- File Management
- Device Management
- Information Maintenance
- Communications

Process Control
This system calls perform the task of process creation, process termination, etc.

Functions:

- End and Abort
- Load and Execute
- Create Process and Terminate Process
- Wait and Signed Event
- Allocate and free memory

File Management
File management system calls handle file manipulation jobs like creating a file, reading, and writing, etc.

Functions:

- Create a file
- Delete file
- Open and close file
- Read, write, and reposition
- Get and set file attributes

Device Management

Device management does the job of device manipulation like reading from device buffers, writing into device buffers, etc.

Functions

- Request and release device
- Logically attach/ detach devices
- Get and Set device attributes

Information Maintenance
It handles information and its transfer between the OS and the user program.

Functions:

- Get or set time and date
- Get process and device attributes

Communication:
These types of system calls are specially used for interprocess communications.

Functions:

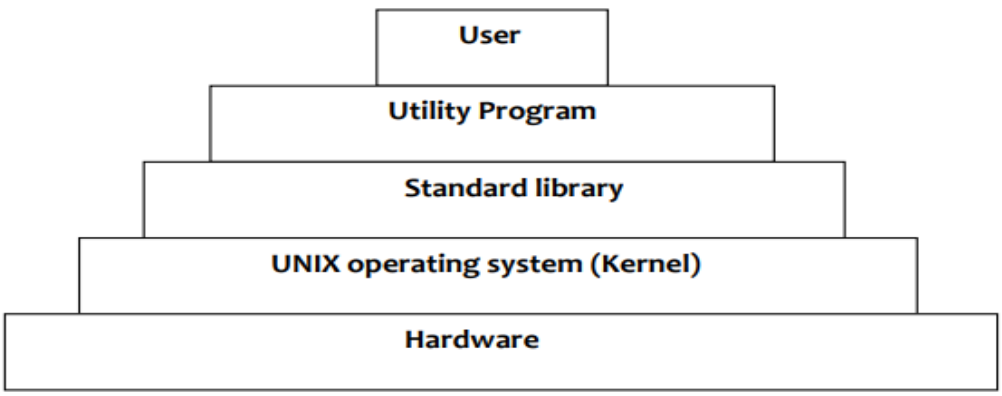
- Create, delete communications connections
- Send, receive message
- Help OS to transfer status information
- Attach or detach remote devices

Examples of Windows and Unix System Calls –

	Windows	Unix
Process Control	CreateProcess() ExitProcess() WaitForSingleObject()	fork() exit() wait()
File Manipulation	CreateFile() ReadFile() WriteFile() CloseHandle()	open() read() write() close()
Device Manipulation	SetConsoleMode() ReadConsole() WriteConsole()	ioctl() read() write()
Information Maintenance	GetCurrentProcessID() SetTimer() Sleep()	getpid() alarm() sleep()
Communication	CreatePipe() CreateFileMapping() MapViewOfFile()	pipe() shmget() mmap()
Protection	SetFileSecurity() InitlializeSecurityDescriptor() SetSecurityDescriptorGroup()	chmod() umask() chown()

Linux Operating system

LINUX is an operating system or a kernel distributed under an open-source license. Its functionality list is quite like UNIX. The kernel is a program at the heart of the Linux operating system that takes care of fundamental stuff, like letting hardware communicate with software.



Linux architecture

Linux Kernel

The main purpose of a computer is to run a *predefined sequence of instructions*, known as a **program**. A program under execution is often referred to as a **process**. Now, most special purpose computers are meant to run a single process, but in a sophisticated system such a general purpose computer, are intended to run many processes simultaneously. Any kind of process requires hardware resources such are Memory, Processor time, Storage space, etc.

In a General Purpose Computer running many processes simultaneously, we need a middle layer to manage the distribution of the hardware resources of the computer efficiently and fairly among all the various processes running on the computer. This middle layer is referred to as the **kernel**. Basically the kernel virtualizes the common hardware resources of the computer to provide each process with its own virtual resources. This makes the process seem as it is the sole process running on the machine. The kernel is also responsible for preventing and mitigating conflicts between different processes.
This schematically represented below:

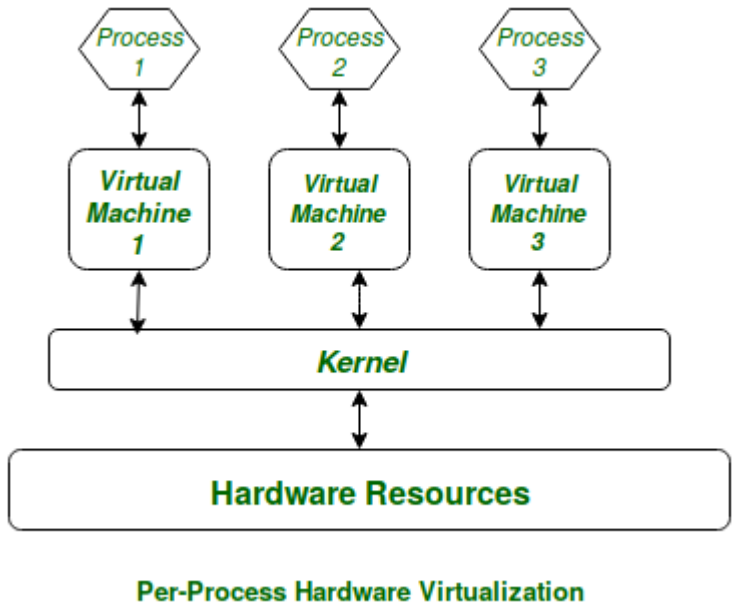


Figure: Virtual Resources for each Process
The **Core Subsystems** of the **Linux Kernel** are as follows:

1. The Process Scheduler
2. The Memory Management Unit (MMU)
3. The Virtual File System (VFS)
4. The Networking Unit
5. Inter-Process Communication Unit

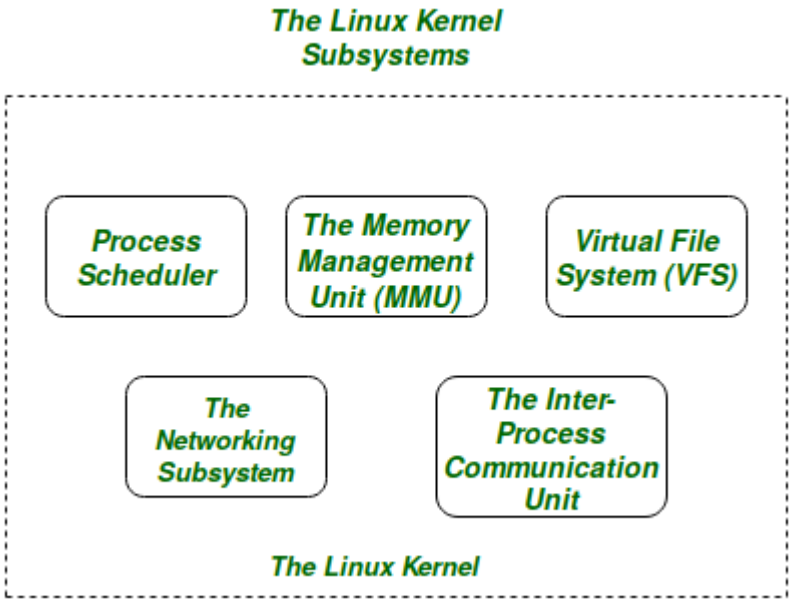


Figure: The Linux Kernel

For the purpose of this article we will only be focussing on the 1st three important subsystems of the Linux Kernel.

The basic functioning of each of the 1st three subsystems is elaborated below:

- **The Process Scheduler:**
  This kernel subsystem is responsible for fairly distributing the CPU time among all the processes running on the system simulteneously.
- **The Memory Management Unit:**
  This kernel sub-unit is responsible for proper distribution of the memory resources among the various processes running on the system. The MMU does more than just simply provide separate virtual address spaces for each of the processes.
- **The Virtual File System:**
  This subsystem is responsible for providing a unified interface to access stored data across different filesystems and physical storage media.

Write a short note on shell

SHELL is a program which provides the interface between the user and an operating system. When the user logs in OS starts a shell for user. It gathers input from you and executes programs based on that input. When a program finishes executing, it displays that program's output.

Shell is an environment in which we can run our commands, programs, and shell scripts. There are different flavors of a shell, just as there are different flavors of operating systems. Each flavor of shell has its own set of recognized commands and functions.

Shell Prompt

The prompt, **$**, which is called the **command prompt**, is issued by the shell. While the prompt is displayed, you can type a command.

Shell reads your input after you press **Enter**. It determines the command you want executed by looking at the first word of your input. A word is an unbroken set of characters. Spaces and tabs separate words.

Types of Shell:

- **The C Shell –**
  Denoted as **csh**
  Bill Joy created it at the University of California at Berkeley. It incorporated features such as aliases and command history. It includes helpful programming features like built-in arithmetic and C-like expression syntax.

  In C shell:

  Command full-path name is /bin/csh,

  Non-root user default prompt is hostname %,

  Root user default prompt is hostname #.

- **The Bourne Shell –**
  Denoted as **sh**
  It was written by Steve Bourne at AT&T Bell Labs. It is the original UNIX shell. It is faster and more preferred. It lacks features for interactive use like the ability to recall previous commands. It also lacks built-in arithmetic and logical expression handling. It is default shell for Solaris OS.

  For the Bourne shell the:

  Command full-path name is /bin/sh and /sbin/sh,

Non-root user default prompt is $,

Root user default prompt is #.

- **The Korn Shell**
  It is denoted as **ksh**
  It Was written by David Korn at AT&T Bell LabsIt is a superset of the Bourne shell.So it supports everything in the Bourne shell.It has interactive features. It includes features like built-in arithmetic and C-like arrays, functions, and string-manipulation facilities.It is faster than C shell. It is compatible with script written for C shell.

  For the Korn shell the:

  Command full-path name is /bin/ksh,

  Non-root user default prompt is $,

  Root user default prompt is #.

Shell Scripts

The basic concept of a shell script is a list of commands, which are listed in the order of execution. A good shell script will have comments, preceded by # sign, describing the steps.

There are conditional tests, such as value A is greater than value B, loops allowing us to go through massive amounts of data, files to read and store data, and variables to read and store data, and the script may include functions.

Example Script

Assume we create a **test.sh** script. Note all the scripts would have the **.sh** extension. Before you add anything else to your script, you need to alert the system that a shell script is being started. This is done using the **shebang** construct. For example –

#!/bin/sh

This tells the system that the commands that follow are to be executed by the Bourne shell. *It's called a shebang because the # symbol is called a hash, and the ! symbol is called a bang*.

To create a script containing these commands, you put the shebang line first and then add the commands –

#!/bin/bash

```
pwd
ls
```

The shell script is now ready to be executed −

```
$./test.sh
```

Upon execution, you will receive the following result −

```
/home/amrood
index.htm  unix-basic_utilities.htm  unix-directories.htm
test.sh    unix-communication.htm    unix-environment.htm
```

Note − To execute a program available in the current directory, use **./program_name**